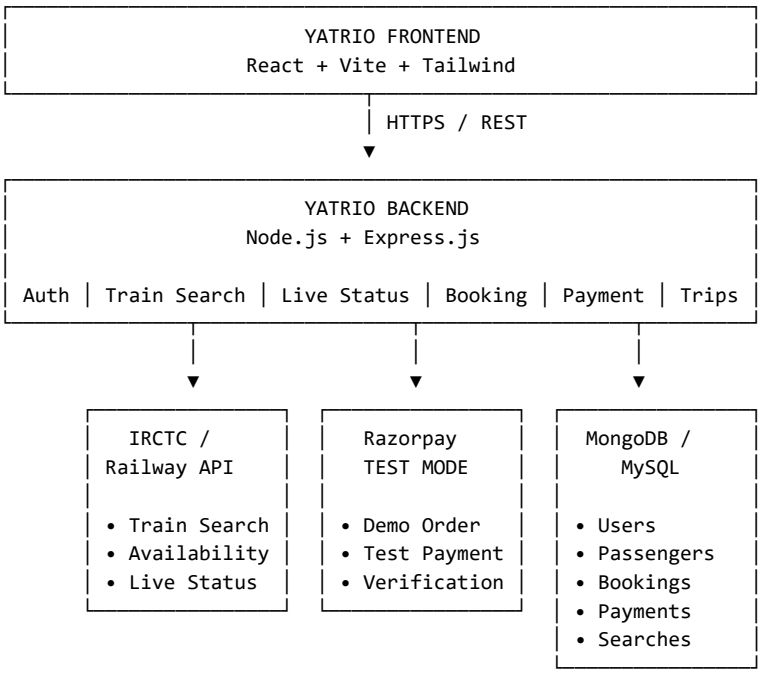
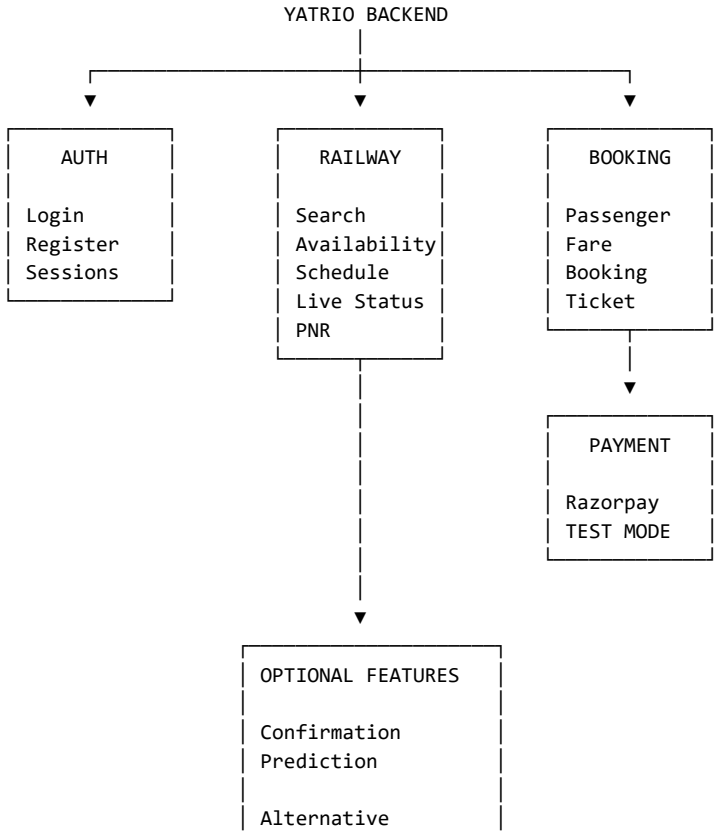
YATRIO

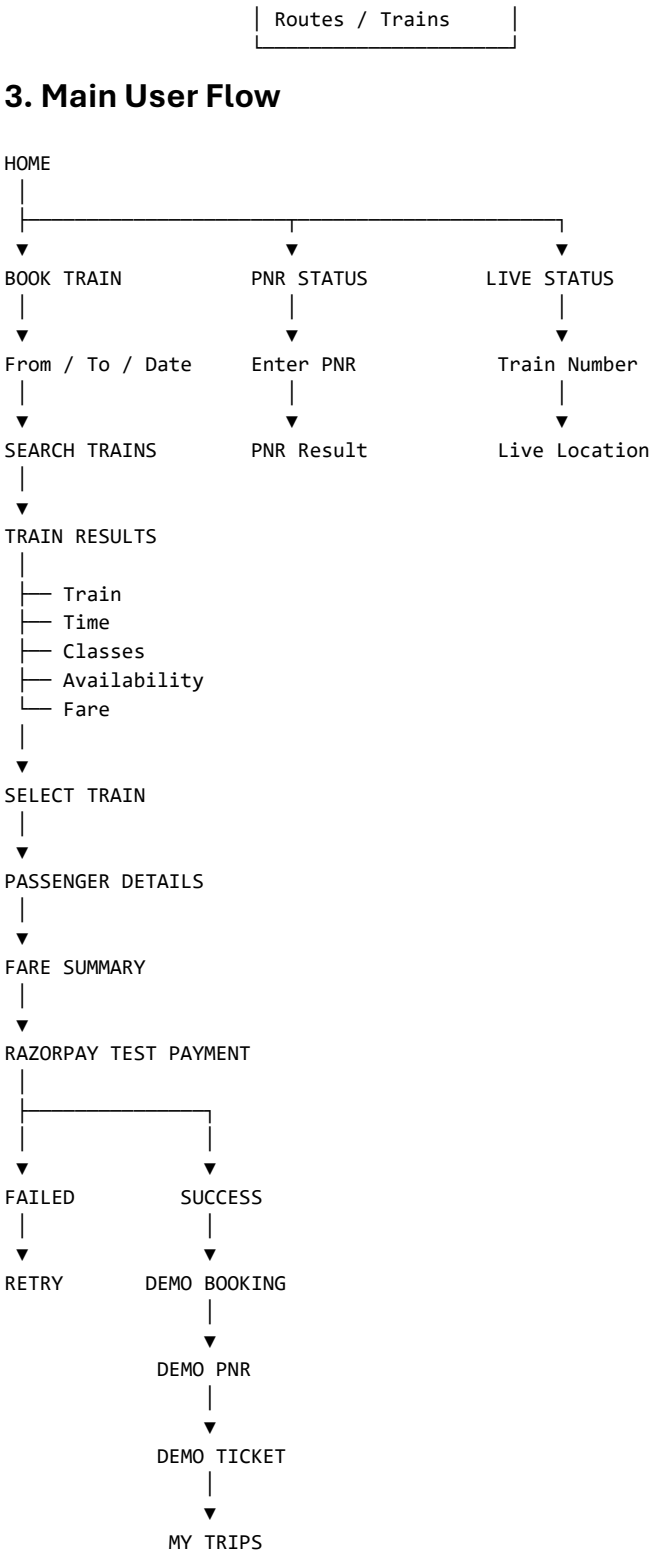
MERN Railway Booking Demo — Architecture & Flows

1. System Architecture



2. Application Modules




Routes / Trains
3. Main User Flow
HOME
BOOK TRAIN
PNR STATUS
LIVE STATUS
From / To / Date
Enter PNR
Train Number
SEARCH TRAINS
PNR Result
Live Location
TRAIN RESULTS
Train
Time
Classes
Availability
Fare
SELECT TRAIN
PASSENGER DETAILS
FARE SUMMARY
RAZORPAY TEST PAYMENT
FAILED
SUCCESS
RETRY
DEMO BOOKING
DEMO PNR
DEMO TICKET
MY TRIPS
4. Train Search Flow
React
Search Form
GET /api/trains/search

```
Express Controller
  │
  ▼
Railway API Service
  │
  ▼
IRCTC / Authorised Railway API
  │
  ▼
Normalize Response
  │
  ▼
React Train Results
```

5. Live Train Location Flow

```
React
  │
  ▼
Train Number
  │
  ▼
GET /api/trains/:trainNo/status
  │
  ▼
Express
  │
  ▼
Railway API Service
  │
  ▼
IRCTC / Railway Live Status API
  │
  ▼
Current Station + Delay + ETA
  │
  ▼
React Live Tracking UI

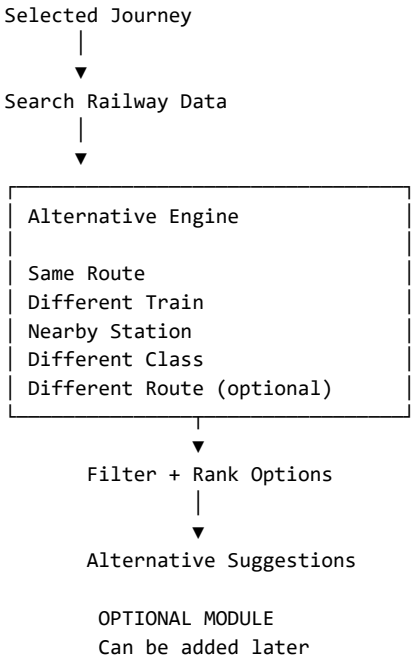
             ┌──────────────────────┐
             │ Optional Redis Cache  │
             │ Latest Status + TTL   │
             └──────────────────────┘
```

6. Optional Confirmation Prediction

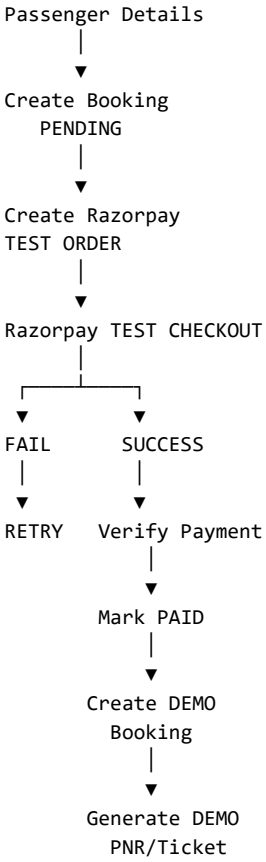
```
TRAIN / AVAILABILITY DATA
          │
          ▼
┌──────────────────────────────┐
│ Confirmation Prediction      │
│                              │
│ • WL / RAC Position          │
│ • Class                      │
│ • Quota                      │
│ • Days Remaining             │
│ • Historical Data (optional) │
└──────────────┬───────────────┘
               ▼
       Estimated Probability
               │
               ▼
        "78% Confirmation"

        OPTIONAL MODULE
        Can be added later
```

7. Optional Alternative Train / Route

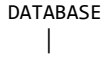


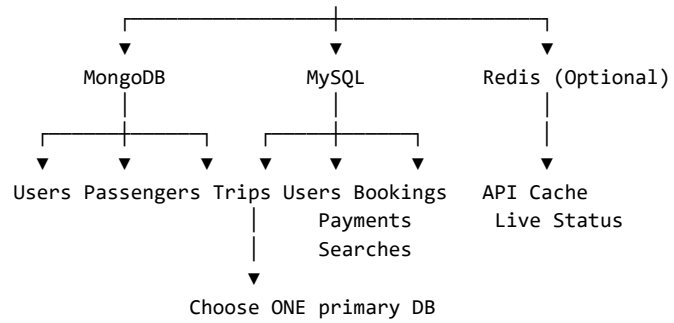
8. Demo Payment + Booking Flow



⚠ NO REAL IRCTC BOOKING

9. Data Architecture





Recommended for MERN:
MongoDB + Mongoose

If relational DB preferred:
MySQL + Prisma/Sequelize

10. Suggested API Map

```
AUTH
POST   /api/auth/register
POST   /api/auth/login

TRAINS
GET    /api/trains/search
GET    /api/trains/:id
GET    /api/trains/:id/availability
GET    /api/trains/:id/schedule
GET    /api/trains/:id/status

PNR
GET    /api/pnr/:pnr

BOOKING
POST   /api/bookings
GET    /api/bookings
GET    /api/bookings/:id
POST   /api/bookings/:id/cancel

PAYMENT
POST   /api/payments/order
POST   /api/payments/verify

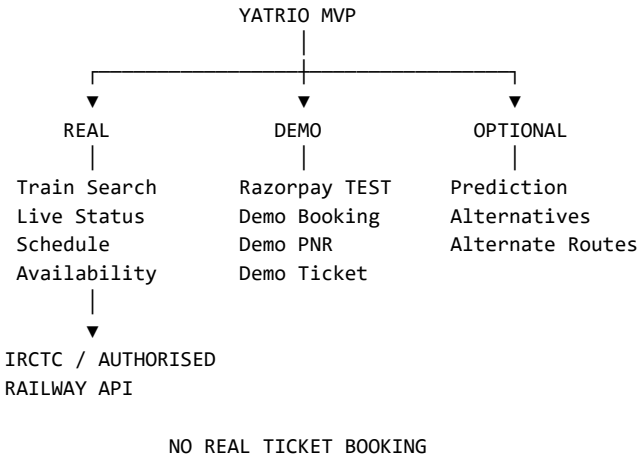
OPTIONAL
POST   /api/prediction
POST   /api/alternatives
```

11. Frontend Pages

```
/                       → Home
/search                  → Train Search
/trains                  → Train Results
/trains/:id              → Train Details
/booking                 → Passenger + Checkout
/booking/success         → Demo Ticket
/trips                   → My Trips
/trips/:id               → Ticket Details
/pnr                     → PNR Status
/live-status             → Live Train Status
/login                   → Authentication

OPTIONAL
/prediction
/alternatives
```

12. Final MVP Boundary



Yatrio • Architecture v1 • MERN + Railway API + Razorpay Test Mode